Worksheet 2.3 — Solving Polynomial Equations

Grade 12 Advanced Functions (MHF4U) · Unit 2: Polynomial Equations and Inequalities

The Big Ideas

Factor fully common factor, grouping, or the factor theorem then set each factor to zero.

- Take out the common factor first.
- Try grouping, or the factor theorem + division.
- A degree-n polynomial has n roots (with multiplicity).

Core Concepts

Factor to solve

Set the polynomial to zero, factor completely using the factor theorem, then apply the zero-product property.

Counting real roots

A degree-n equation has at most n real roots; irreducible quadratic factors contribute non-real roots instead.

Repeated roots

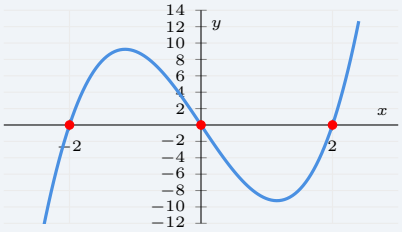
A repeated factor produces a repeated root, seen on the graph as a touch or a flattening at the axis.

Worked Examples

Example 1 — Common factor

Solve $3x^3 - 12x = 0$.

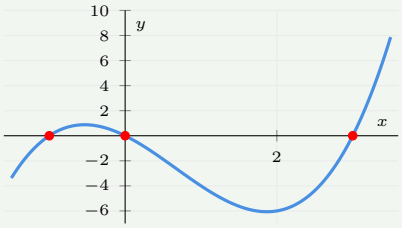
Solution. $3x(x-2)(x+2) = 0 \Rightarrow x = 0, 2, -2$. The graph crosses at all three:



Example 2 — Factor a trinomial

Solve $x^3 - 2x^2 - 3x = 0$.

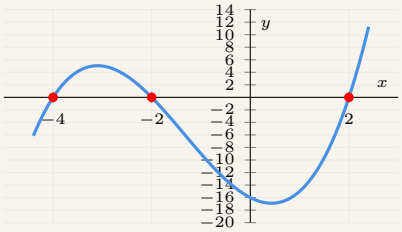
Solution. $x(x-3)(x+1) = 0 \Rightarrow x = 0, 3, -1$:



Example 3 — Grouping

Solve $x^3 + 4x^2 - 4x - 16 = 0$.

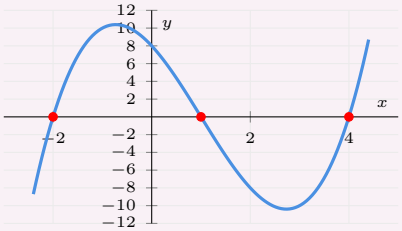
Solution. $x^2(x+4) - 4(x+4) = (x+4)(x-2)(x+2) = 0 \Rightarrow x = -4, 2, -2$:



Example 4 — Factor theorem

Solve $x^3 - 3x^2 - 6x + 8 = 0$.

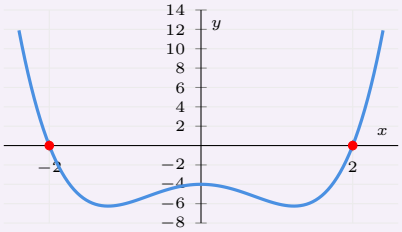
Solution. $P(1) = 1 - 3 - 6 + 8 = 0$; divide $(x-1)(x-4)(x+2) = 0 \Rightarrow x = 1, 4, -2$:



Example 5 — Quadratic in x^2

Solve $x^4 - 3x^2 - 4 = 0$.

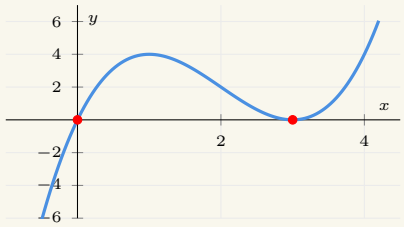
Solution. $(x^2-4)(x^2+1) = 0$; $x^2 + 1 \neq 0$ for real x, so $x = \pm 2$:



Example 6 — Repeated factor

Solve $x^3 - 6x^2 + 9x = 0$.

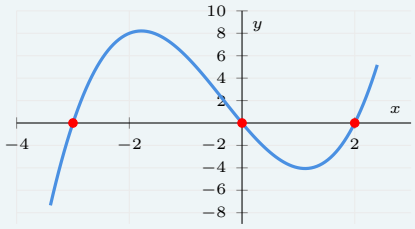
Solution. $x(x^2 - 6x + 9) = x(x-3)^2 = 0 \Rightarrow x = 0, 3$ (a double root at 3, where the graph touches):



Example 7 — Trinomial

Solve $x^3 + x^2 - 6x = 0$.

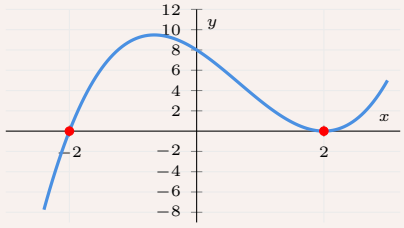
Solution. $x(x+3)(x-2) = 0 \Rightarrow x = 0, -3, 2$:



Example 8 — Grouping

Solve $x^3 - 2x^2 - 4x + 8 = 0$.

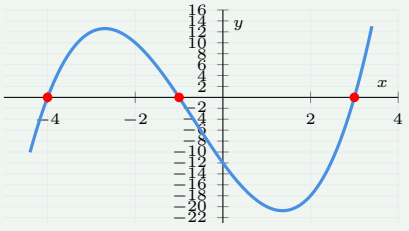
Solution. $x^2(x-2) - 4(x-2) = (x-2)(x-2)(x+2) = 0 \Rightarrow x = 2$ (double), $x = -2$:



Example 9 — Factor theorem

Solve $x^3 + 2x^2 - 11x - 12 = 0$ (hint $x = -1$).

Solution. $(x+1)(x+4)(x-3) = 0 \Rightarrow x = -1, -4, 3$:



Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Solve $x^3 - 49x = 0$.

Question 2

Solve $x^3 + x^2 - 12x = 0$.

Question 3

Solve $x^3 - 5x^2 - x + 5 = 0$ by grouping.

Question 4

Solve $x^3 - x^2 - 14x + 24 = 0$ (hint $x = 2$).

Question 5

Solve $x^4 - 5x^2 - 36 = 0$.

Question 6

Solve $x^3 - 16x = 0$.

Question 7

Solve $x^3 - 3x^2 - 4x = 0$.

Question 8

Solve $x^3 - x = 0$.

Question 9

Solve $x^4 - 13x^2 + 36 = 0$.

Question 10

Solve $x^3 + x^2 - 9x - 9 = 0$ by grouping.

Question 11

Solve $x^3 - 2x^2 - 5x + 6 = 0$ (hint $x = 1$).

Question 12

Solve $2x^3 - 8x = 0$.

Question 13 — Challenge

Solve $x^3 - 6x^2 + 11x - 6 = 0$ (hint: try $x = 1$).

Question Recap & Answer Key

Question Recap

1. Solve $x^3 - 49x = 0$.
2. Solve $x^3 + x^2 - 12x = 0$.
3. Solve $x^3 - 5x^2 - x + 5 = 0$ by grouping.
4. Solve $x^3 - x^2 - 14x + 24 = 0$ (hint $x = 2$).
5. Solve $x^4 - 5x^2 - 36 = 0$.
6. Solve $x^3 - 16x = 0$.
7. Solve $x^3 - 3x^2 - 4x = 0$.
8. Solve $x^3 - x = 0$.
9. Solve $x^4 - 13x^2 + 36 = 0$.
10. Solve $x^3 + x^2 - 9x - 9 = 0$ by grouping.
11. Solve $x^3 - 2x^2 - 5x + 6 = 0$ (hint $x = 1$).
12. Solve $2x^3 - 8x = 0$.
13. Solve $x^3 - 6x^2 + 11x - 6 = 0$ (hint: try $x = 1$).

✓ Answer Key

1. $0, 7, -7$
2. $0, -4, 3$
3. $5, 1, -1$
4. $2, -4, 3$
5. ± 3
6. $0, 4, -4$
7. $0, 4, -1$
8. $0, 1, -1$
9. $\pm 2, \pm 3$
10. $-1, 3, -3$
11. $1, 3, -2$
12. $0, 2, -2$
13. $1, 2, 3$